SentinelQ Scoring Pipeline — Complete Reference

v2.0.0 · 2026-10-09 · worked example: Bajaj Finance (BAJFINANCE), from the 2026-10-09 run

Overview

SentinelQ scores each stock three separate ways through six stages. Only one stage uses AI; everything else is deterministic arithmetic reading a fixed rubric file, so any number here can be recomputed by hand.

What changed in v2.0.0: Sentiment dropped its x50 scaling and now shows a Raw mean (plain average) and a Weighted mean (recency x confidence) side by side, both banded directly. Corporate Actions was rebuilt entirely on a severity x stage x timing x size model (replacing the old base x materiality clip), combined as a single 0-10 score — not two. An article can now feed Sentiment, Corporate Actions and Governance all at once, where before the search pass that found it gated which score(s) it could touch.

Stage	What happens	AI involved?
1. Input	Read the stock list, resolve name/sector/aliases	No
2. Ingest	Fetch news + corporate actions, filter obvious junk, cluster duplicates into events	No
3. Classify	Label each event: type, sentiment -2..+2, governance flag, and (new) Corporate-Action stage/size/timing	Yes — the only AI step
4. Validate	Schema-check every label; retry once, then drop with a reason	No
5. Score	Turn labels into Sentiment (raw+weighted), Governance, and Corporate Actions, via the rubric	No
6. Report	Write the Scorecard, PDF, and supporting evidence sheets	No

Stage 1 — Input

Reads a 3-column list (symbol, company, sector); resolves sector/cap/aliases from a reference file. Unchanged in v2.0.0, with one fix: a company's own registered-name words now always feed the matching-token list, even when an alias list is also configured — previously an alias list silently replaced the name's own words, which could drop genuine articles before the model ever saw them.

Why: *every later search depends on knowing exactly which name/ticker/aliases identify the company.*

Stage 2 — Ingest

Three separate 12-month searches per stock — general news, a dedicated results search, a dedicated governance-keyword search — plus a structured corporate-actions feed. Unchanged in v2.0.0.

The pre-AI filter bar is lighter in v2.0.0. The title-match check is unchanged (a headline must still name the company). The templated-headline filter was trimmed: record-date/ex-dividend notices, “in focus”, “here's why”, “block/bulk deal”, “multibagger” and similar phrasing are no longer blind-filtered before the model reads them — a single-company block-trade notice or a dividend-date article can now reach the model instead of being discarded on a regex alone. What stays filtered: formats that are structurally multi-stock roundups (“Stocks to

watch: X, Y, Z", "Dividends: Last Day To Buy" listing several names) — these carry no single-company content by construction, so filtering them loses nothing.

***Why:** a blanket regex can't tell a genuine single-company notice from a listicle that happens to share its wording; the model, now reliably good at judging substance, is a better filter for anything that's ambiguous. The clustering/dedupe step that merges duplicate coverage into one event is unchanged either way.*

The per-day cap and the 40/25/20/15 window quotas are unchanged, and — as before — only apply when a run is explicitly started with `--budget` / `--window-quotas`. The Bajaj Finance run below used neither: 367 articles retrieved, all of them usable were sampled.

Stage 3 — Classify: the AI's rubric

The sentiment anchor table (-2..+2) is unchanged from the prior version. What's new in v2.0.0 is the taxonomy the model chooses from, and three optional fields it now fills for corporate-action events.

Score	Criteria
+2	Material, company-specific, hard-fact good news: growth ≥ ~20% or a clear beat, guidance raised, a large order/capacity add, a rating upgrade, a broker target raise ≥ ~15%, a clean regulatory chit
+1	Positive but modest/routine: single-digit growth, in-line quarter, small order, reiterated Buy, ordinary dividend
0	No value signal: passing mention, a list, price-only item, routine filing, neutral broker view
-1	Mildly negative: soft quarter, small miss, downgrade to neutral, modest fine/demand, unplanned non-CXO exit
-2	Material bad news: profit fall ≥ ~20% or a big miss, guidance cut, regulatory order/settlement, fraud/probe, auditor exit, CXO resignation

Taxonomy expanded from 30 to 45 event types (see Section 10 for the full list), adding the full Corporate Action Scoring Rules taxonomy: insolvency_admission, demerger, open_offer_takeover, rights_issue, pledge_invocation, special_dividend, lock_in_expiry, name_symbol_isin_change, fy_end_change, routine_agm_egm, admin_notice, noise_notice, index_inclusion, index_exclusion, exchange_surveillance.

Three new optional fields feed the Corporate Actions formula (Stage 5c): `ca_stage` (board_intimation / board_approved / approval_obtained / completed), `ca_size_pct` (dilution %, pledged % of promoter holding, or rights discount %, where stated), `ca_days_to_effective` (days from publish date to a stated future effective/record date). All optional — the formula has a sensible default for each when the text doesn't say.

***Why:** the prompt revision that added these bumped PROMPT_VERSION to v3, so every article gets freshly labelled against the new schema rather than silently reusing a pre-v3 label that never had these fields.*

The four cage bars and Stage 4 — Validate are unchanged from the prior version: forced JSON schema, extraction only, a label cache standing in for true temperature-0, and full response logging; schema failures retry once then drop with a reason, and >10% label failures halts the run (`--allow-partial-labels` to override).

Stage 5a — Sentiment scoring (rewritten in v2.0.0)

No x50 scaling anywhere. The prior version computed a -2..+2 mean, stretched it onto a -100..+100 “points” scale (mean x 50), and banded the points. v2.0.0 drops that step entirely: the mean is banded directly on the -2..+2 scale it's already on.

Two means, not one. The Weighted mean applies a recency weight and (optionally) a source-confidence weight, exactly as before. A new Raw mean sits beside it — the plain, unweighted average of the same counted articles, no recency, no confidence. Both are shown, both are banded the same way, and only the Weighted one is “the” headline Sentiment score.

```
weighted_mean = Σ(w_i × s_i) / Σ w_i raw_mean = Σ s_i / n (w_i = recency weight; confidence is OFF by default, same as before)
```

Window weight (unchanged): age looked up against the same fixed boundaries as before.

Age	Window	Weight
0-30 days	W1	1.00
30-90 days	W2	0.75
90-180 days	W3	0.50
180-365 days	W4	0.25

Why: *fresh news dominates (W1 at full weight), but a genuinely bad year isn't erasable just because the bad news happened early — W4 still carries a quarter of full weight, never zero.*

Band cutoffs (new) — both means use the same bands:

Band	Mean on the -2..+2 scale
+2	≥ 1.6
+1	0.5 to 1.59
0	-0.49 to 0.49
-1	-1.59 to -0.5
-2	≤ -1.6

Why: *reaching +2/-2 now needs real consensus (≥ 1.6, not a bare round-to-nearest) — a mean of 1.2 stays +1. This replaced the old points-based bands (≥50 pts → +2) when the x50 step was removed; the ratio is the same shape, just expressed directly on the native scale.*

Which articles count — unchanged in spirit, widened in scope. The exclusion reasons from the prior version (price-only items, passing mentions, a zero-sentiment label on a fixed administrative-type list) are all unchanged. What changed: in the prior version, only articles from the dedicated sentiment-sample search could feed the mean — an article found purely by the governance-keyword search was walled off from sentiment even if it described a real, sentiment-relevant event. v2.0.0 removes that wall: any news article with a real event about the company can feed the mean, regardless of which search found it.

Why: *an auditor resignation found only by the governance search is still genuine evidence of how the year went — it shouldn't be invisible to sentiment just because of which search happened to surface it. This is the same change that lets one article feed Sentiment, Corporate Actions and Governance all at once.*

Worked example — Bajaj Finance, this run

	Decimal mean	Banded score
Weighted mean	+0.786	+1
Raw mean	+0.774	+1

56 relevant articles counted, evidence span 337 days of the 12-month lookback. Both means land in the same +1 band here (0.5 to 1.59) — when they diverge more, it's a signal that recent news is telling a different story than the full-year average.

Stage 5b — Governance scoring

Starts at 100. Every governance-flagged item passes the same verification gate as before (subject, occurred-at-company, severity, action-stage) before it can cost anything. Unchanged structurally in v2.0.0; two additions to the penalty table, both drawn from the new Corporate Actions taxonomy so the same event can be penalised under Governance and scored under Corporate Actions from the one article:

Event	Penalty	Why
pledge_invocation	-20	Forced selling into the market — the classic mid/small-cap stress spiral
pledge (creation)	0 / -5 / -10	Penalised only at medium/high materiality — pledging itself isn't necessarily adverse
insolvency_admission	-25	Can go to near zero and may not be exitable

Labels: Clean ≥95 · Watch ≥80 · Flag below. Age taper: full penalty for 182 days, fading linearly to 50% by 365 days. Both unchanged.

Worked example — Bajaj Finance, this run

13 items flagged, 0 penalised → governance stays at 100, Clean.

Date	Headline	Event type
2025-11-12	Jennifer Mathew Fernandes elevated to CHRO, Bajaj Finance	management change routine
2025-11-18	Jennifer Mathew Fernandes Appointed as Chief Human Resourc	management change routine
2026-01-29	Delhi HC restrains “SettlementGuru” from using Bajaj Finan	litigation
2026-03-17	Bajaj Finance Appoints Tushar Deb as Senior Management Per	management change routine
2026-04-29	Rajiv Bajaj to step down as Bajaj Finance director after e	management change routine
2026-04-29	Bajaj Finance Q4 PAT rises 22% to ■5,465 crore; lower prov	earnings beat
2026-04-30	Stock Alert: Adani Power, Force Motors, Hindustan Unilever	board change routine
2026-06-10	Rajiv Bajaj to step down as Non-Executive Director of Baja	board change routine

Stage 5c — Corporate Actions scoring (rebuilt in v2.0.0)

The prior version scored each corporate action on a fixed base value (dividend +1.0, buyback +1.5, split/bonus +0.5, dilution -1.5...) x a materiality multiplier, summed and clipped to -2..+2. v2.0.0 replaces this entirely with the severity x stage x timing x size model from the Corporate Action Scoring Rules document (8-Oct-2026), covering the full 20-type severity table rather than 5 types.

severity x stage x timing x size, capped at 10 per event; Risk and Opportunity combined separately as 1 - product(1 - event_score/10) (chance at least one bites), never summed or averaged

Step 1 — Severity (the event type's ceiling, 0-10)

Severity	Event type	Damage
10	insolvency admission	both
9	demerger	both
9	merger acquisition	both
8	split bonus	data
8	open offer takeover	economic
7	rights issue	both
7	pledge	economic
7	pledge invocation	economic
6	dilution	both
6	special dividend	both
6	auditor resignation	economic
6	management exit	economic
5	buyback	economic
5	lock in expiry	economic
4	dividend	data
3	name symbol isin change	data
2	fy end change	data
2	routine agm egm	data
1	admin notice	none
0	noise notice	none

Damage: *Economic* = the company is genuinely worth less (or more) — real value or real risk changed. *Data* = nothing about value changed, but price/fundamentals data breaks if the adjustment factor isn't applied everywhere at once (a 1:4 bonus isn't a loss, but an unadjusted price series shows an 80% crash that never happened). *Both* = carries real economic consequence AND will corrupt unadjusted data (a demerger: the parent really is worth less, and the price history shows a cliff if not adjusted).

Step 2 — Stage multiplier (how certain is it)

ca_stage	Multiplier	Meaning
board_intimation	0.3	Board informed / just announced
board_approved	0.6	Board has approved it

ca_stage	Multiplier	Meaning
approval_obtained	0.8	Shareholder or regulatory approval obtained
completed	1	Record date fixed or already happened

If the label doesn't state a stage, it defaults to `board_approved` — a deliberate middle value.

Step 3 — Timing multiplier (how close or how stale)

Upcoming (the label states a future effective/record date): banded by how far out.

Days to effective	Multiplier
< 15 days	1
15-60 days	0.75
> 60 days	0.5

Already happened: decays as 0.5 ^ (days_old / half_life) — by definition, at exactly one half-life the multiplier is 0.5; it approaches zero but never reaches it.

Half-life	Event types	Why
30 days	dividend, buyback, special_dividend	A one-day cash event — a month later it's genuinely old news
90 days	merger_acquisition, demerger, open_offer_takeover (and the default for every other type)	Matters longer, decays slower
180 days	pledge, pledge_invocation, auditor_resignation, management_exit	Describes an ongoing state, not a single-day event — stays live longer

Step 4 — Size multiplier (how big is this instance)

ca_size_pct	Multiplier
≤ 2%	0.7
≤ 5%	1
≤ 10%	1.2
≤ 999%	1.5

Applies to dilution %, pledged % of promoter holding, or rights discount %, where the label states one. If not, materiality stands in: low → 0.7, medium → 1, high → 1.5. If neither applies to the event type (an admin notice, an auditor exit), the multiplier is 1 — no scaling, not a penalty for missing data.

Step 5 — Multiply, cap at 10

event_score = min(10, severity × stage × timing × size). The cap exists because the raw product can overshoot (e.g. insolvency_admission 10 × completed 1.0 × fresh 1.0 × largest size 1.5 = 15) — once the arithmetic says “already maximal”, a larger number adds no information, so it's clamped to the scale's own ceiling.

Step 6 — Direction (tracked separately, never multiplied in)

label.sentiment > 0 -> positive (feeds Opportunity); < 0 -> negative (feeds Risk); 0 -> unclear (feeds Risk, as the table specifies 'negative or unclear')

Why: *the same real event can be read two ways by two different articles — e.g. a capital raise is dilutive (negative for existing holders) and promoter-backed (positive, signals confidence) at once. The severity x stage x timing x size arithmetic only answers “how big a deal, mechanically”; direction is a separate judgement applied after, to sort the event into the Risk or Opportunity pool.*

Step 7 — Combine: one score, not an average or a sum

Risk pools every negative + unclear-direction event score; Opportunity pools every positive one. Each combines as:

```
score = 10 × [ 1 − ∏ (1 − event_score_i / 10) ] — “the chance at least one bites”
```

Summing would let unrelated minor events stack past any sensible ceiling. Averaging would let weak events dilute a serious one. Treating each score as a probability out of 10 avoids both: one dominant event drives the number, minor ones nudge it up with diminishing effect, and it can never exceed 10 no matter how many events pile on.

The Corporate Actions score shown everywhere is the Risk pool's result — not a Risk/Opportunity pair. Opportunity is still computed in full (from the positive-direction pool, same formula) but only ever surfaces as a separate “event-trade review” flag when it crosses 6, and it never changes the score or band above it.

***Why:** the rules document is explicit: “good news never cancels bad news”, and “when both scores are high, the Risk band governs.” A strong Opportunity reading is worth a look for a different kind of review (tender into a buyback, hold through an open offer) — it is never a reason to downgrade the Corporate Actions score.*

Step 8 — The two gates

Gate	Triggers on	Effect
Gate A — hard override	insolvency_admission, pledge_invocation; auditor_resignation only when the label's severity is ‘integrity’ (withheld-information/irregularity, not a routine explained exit)	Forces the Exit band regardless of the arithmetic — no weighted score should argue someone out of exiting a company that just admitted insolvency.
Gate B — data quarantine	split bonus, demerger, merger acquisition, rights issue, special dividend, name symbol isin change	Excluded entirely before the combine step — these carry no real economic risk; their only effect is corrupting unadjusted price/fundamentals data, which the rules doc's adjustment-factor checks handle separately.

A third bucket, **tracked but not scored**: index inclusion, index exclusion, exchange surveillance — kept on the calendar and shown, but excluded from the arithmetic entirely; the rules doc treats these (and results announcements, already scored under Sentiment) as disclosure events, not formal corporate actions.

Step 9 — Band

Score	Band
0–2.9	Log
3–4.9	Watch
5–6.9	Review
7–8.9	Act
9–10	Exit

Step 0 (applies throughout) — Dedupe

Repeat headlines of one declaration are ONE action: dividends within 120 days, everything else within 30 days, collapse to a single event before scoring. This is distinct from the ingest-stage clustering in Stage 2 — this dedupe runs specifically on labelled corporate-action events, catching cases the earlier clustering didn't fully merge.

Worked example — Bajaj Finance, every Corporate Actions event this run

Event	Sev.	Stage (x)	Timing (x)	Size (x)	Score	Direction
dilution	6	board_approved (0.6)	0.97 (4d old, half-life 90d)	1	3.49	negative
dilution	6	board_approved (0.6)	0.948 (7d old, half-life 90d)	1	3.41	positive
dilution	6	board_intimation (0.3)	0.891 (15d old, half-life 90d)	1	1.60	unclear
dividend	4	board_approved (0.6)	0.1 (102d old, half-life 30d)	1	0.24	positive
dividend	4	board_approved (0.6)	0.1 (101d old, half-life 30d)	1	0.24	positive
admin notice	1	board_approved (0.6)	0.241 (185d old, half-life 90d)	1	0.14	unclear

Risk pool [3.49, 1.6, 0.14]: 1 – (1-3.49/10) × (1-1.60/10) × (1-0.14/10) = 1 – 0.539 = 0.461 × 10 = **4.61** → Watch

Opportunity pool [3.41, 0.24, 0.24]: 1 – (1-3.41/10) × (1-0.24/10) × (1-0.24/10) = 1 – 0.628 = 0.372 × 10 = 3.72 < 6 → no review flag

Gate B quarantined: **split bonus** (2026-06-16, score 1.98) — excluded from the pools above entirely, contributes nothing to either number.

Gate A: did not fire this run — nothing here was insolvency, pledge invocation, or a withheld-information auditor exit.

Stage 6 — Report

Writes the Scorecard, PDF, and supporting Excel sheets — two new sheets and two new Scorecard columns in v2.0.0.

Sheet	What's there
Read me	Plain-English glossary of every column and band
Scorecard	Per-stock Sentiment (Weighted + Raw, each with a decimal-mean column alongside the banded score), Governance score/label, Corporate Actions score/band/event-trade-review flag
Articles by window	One row per article ever found — 'How it was treated' spells out the exact reason, scored or not
Window summary	Per-window counts (+2..-2), weighted push added, full-text-read count
Evidence - Governance	Every flagged item: penalty applied, or why not, with n_sources/tier/confidence
Corporate Actions (new)	One row per stock: score, band, event-trade-review flag, Gate A/B hits, counts
Corporate Actions - Events (new)	Every scored/gated/tracked event: full severity×stage×timing×size arithmetic, direction, and a Sourcing column (source, tier, source count, confidence) flagging any single lower-tier source in red — treat its date and facts as unconfirmed
Coverage	Retrieved/kept/dropped counts per stock, plus the full dropped-items list with reasons
Run Info	Rubric version/hash, as-of date, retrieval mode

Why: *every number above should be checkable by hand against a real row, not taken on faith — the new Sourcing column exists specifically so a weakly-sourced date or fact is visible, not silently presented as settled.*

rubric_v1.json — what changed (field reference)

Version **2.0.0** (was 1.9.1). The file's own words: “Single source of truth for every weight, penalty and taxonomy entry. Scores change only if this file changes.”

Sentiment block — new/changed fields

Field	Value	Note
display_scale	raw_-2_to_2_no_x50	No x50 anywhere; both means banded on -2..+2 directly
band_cutoffs	plus2_at 1.6, plus1_at 0.5, minus1_at -0.5, minus2_at -1.6	The live banding rule
points_scale, bands (old)	50 / {plus2:50, plus1:15}	Dormant — kept only for any code path still reading sentiment_points

Corporate actions block — entirely new structure

Replaces: clip [-2,2], base {5 types}, materiality_multiplier. Now: severity {20 types}, damage_type, stage_multiplier, timing{upcoming bands, past_half_life_days}, size_bands, size_fallback_from_materiality, direction_from_sentiment, gate_a_hard_exit, gate_b_data_quarantine, bands {5 bands}, opportunity_review_at, disclosure_tracked_not_scored. Full values given stage by stage above (Stage 5c). dedupe is unchanged (dividend 120d / other 30d).

event_types — 45 (was 30)

All 30 prior types retained; 15 added: insolvency_admission, demerger, open_offer_takeover, rights_issue, pledge_invocation, special_dividend, lock_in_expiry, name_symbol_isin_change, fy_end_change, routine_agm_egm, admin_notice, noise_notice, index_inclusion, index_exclusion, exchange_surveillance.

Governance block

Unchanged except the three new penalty entries in Stage 5b above (pledge, pledge_invocation, insolvency_admission).

config/__init__.py — what changed

Field	Value	Note
BOILERPLATE_PATTERNS	~14 patterns (was 31)	Trimmed to formats mechanically certain to carry no company event (live price trackers, 52-week markers, market-wide recaps, pure PR filler) or structurally multi-stock (listicles, dividend-date roundups). See Stage 2 for the full before/after reasoning.
NOISE_ZERO_TYPES	unchanged	{other, management_change_routine, board_change_routine, dilution, rpt_routine} — still excluded from the sentiment mean when labelled 0, regardless of which pass found them
PER_DAY_CAP, SAMPLE_WINDOWS	unchanged (6/day; 40/25/20/15)	Still only enforced with --budget / --window-quotas

Multi-label articles — the architectural change underlying all of the above

Before v2.0.0, an article's eligibility for Sentiment vs Governance was gated by `item.purpose` — set once, before the AI ever ran, based on which search pass retrieved it. A governance-pass article could only ever affect the governance score, even if it was obviously also sentiment-relevant news.

v2.0.0 decouples this. Scoring eligibility for each of the three scores is now computed independently, from the article's own label content (event type, substance, sentiment, governance_flag) — never from which search found it. The practical effect: one article — say, an auditor resignation — can now feed Sentiment (it's bad news), Corporate Actions (auditor_resignation has severity 6), and Governance (a fixed -20 penalty) all from a single read.